

MATH-619

A CNN-LSTM Hybrid Model for Multivariate Stock Price Forecasting Using Technical and Sentiment Indicators

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1. Introduction and Problem Definition

1.1 General Area of Study

Financial markets produce massive amounts of data every trading day. There are numerous types of financial market data including stock price, trading volume, technical indicators, news sentiment, and macro-economic variables; however, there is no singular analytical method to account for the simultaneous interaction of all these variables. Since the late 1950's, quantitative traders, hedge fund managers and academic researchers have attempted to develop methodologies that can identify predictive signals within the complexities of financial data and translate them into directional forecasts: Will a given stock increase in value or decrease in value tomorrow compared to its current price?

This inquiry represents the confluence of financial economics and applied data science. The traditional solution based on the Efficient Market Hypothesis (EMH) of Fama (1970) posits that all publicly available information about assets are priced-in by the market and thus impossible to predict. Despite EMH, decades of behavioral finance and market microstructure research have provided evidence of numerous market anomalies - momentum effect, short-term reversals, sentiment-based mis-pricing, and volatility clustering - which indicate that the price of securities do not reflect all information instantaneously and perfectly. Furthermore, the magnitude of these anomalies is largest at shorter horizons and in individual equities where most data-driven predictive models are utilized.

Prior to the advent of deep learning techniques, early computational approaches to predicting future directions of stock prices employed classical statistical methods such as ARIMA and linear regression. Although these models are elegant they possess significant limitations. They are predicated upon the assumption that relationships between input variables and output variables are linear in nature and that the data is stationary - assumptions that financial returns commonly violate. Therefore, the signal contained in market data is non-linear and distributed sequentially through time in complex patterns that linear models are unable to capture.

The emergence of deep learning has transformed this field. Neural networks capable of learning non-linear relationships between input variables and output variables from raw data, while retaining memory for long sequences of data, provide a significantly different class of tools for analyzing financial time-series data. Long Short-Term Memory (LSTM) networks in particular - specifically designed to utilize sequence dependencies without experiencing the vanishing gradients associated with prior recurrent architectures - appear to be an effective choice for financial applications. Convolutional Neural Networks (CNNs), originally created for image recognition purposes, have also been successfully adapted to extract spatially localized patterns from financial sequences and therefore capture short-range momentum and reversal effects better than recurrent architectures. It appears reasonable to consider the utilization of both approaches together in hybrid CNN-LSTM architectures - utilizing CNN layers to detect localized patterns in the sequences, followed by LSTM layers that utilize those patterns to create a sequential forecast.

Another developing area of research utilizes natural language processing (NLP) and sentiment analysis to analyze financial data. Financial analysts and investors use social media posts, news articles, analyst reports and earnings calls to gather information about public perception regarding various investment opportunities. As such, if we can extract and quantify this sentiment signal it may enhance predictive capabilities beyond those achievable using only technical price-volume indicators. A key issue facing researchers attempting to combine sentiment with deep learning-based forecasting frameworks involves determining the optimal methodology to include sentiment within these frameworks - specifically whether sentiment enhances incremental predictive capability relative to technical indicators alone, and whether attention mechanisms are required to weigh relevant time periods of sentiment.

1.2 Specific Topic of Study

This project aims to investigate a specific and well-defined prediction task: forecasting the price-direction of a portfolio consisting of five large-cap U.S. technology companies for each subsequent trading day (i.e., whether each company will experience a price increase (UP) or decrease (DOWN)). The five companies are Apple (AAPL), Amazon (AMZN), Alphabet (GOOG), Microsoft (MSFT), and Tesla (TSLA).

These five companies were selected for three intentional reasons. First, these five companies are among the most liquid stocks globally and thus ensure high quality and full historical price data with minimal issues related to liquidity-induced anomalies. Second, technology stocks tend to exhibit greater price volatility than many other sectors providing more variability in directional changes for a model to learn from and making the prediction task more representative of actual trading conditions. Lastly, and most critically for this study, these five companies receive extraordinary levels of social media and financial news coverage. Thus, the social media and news coverage received by Apple, Amazon, Google, Microsoft, and Tesla offers extremely rich sentiment data - which is essential for conducting sentiment analyses as part of this study.

To examine the incremental contributions of three distinct sets of information sources (raw price/volume data alone, technical indicators combined with raw price/volume data, and sentiment-enriched data combined with technical indicators), this project examines three progressively richer feature sets. The feature sets examined are: Condition C1: Raw price/volume data (Open/Close/High/Low/Volume); Condition C2: Eleven pre-computed technical indicators derived from price and volume data - two Exponential Moving Averages (EMA_20, EMA_50), the Relative Strength Index and its signal line (RSI, RSI_MA), the MACD oscillator and its signal line and histogram (MACD, Signal, Histogram), Bollinger Bands (BB_Basis, BB_Upper, BB_Lower), and the CBOE Volatility Index (VIX); and Condition C3: Three feature sets: Condition C1, Condition C2, and Condition C3 includes all features in Conditions C1 & C2, plus one additional feature set containing a daily sentiment score derived from social media and financial news covering four of the five stocks (Microsoft is excluded from C3 because its sentiment series was found to be identical to Amazon's, with a Pearson correlation coefficient of $r = 1.000$, indicating a data provenance issue in the Kaggle source dataset).

This three-condition experimental design enables this project to determine whether the inclusion of technical indicators over raw price/volume data improves predictive capability beyond 50%, and whether the addition of sentiment-enriched data improves predictive capability beyond technical indicator-only models.

1.3 Problem Statement and Significance

Although the growing number of research projects examining deep learning techniques for financial forecasting offer significant potential for improving predictive accuracy and reducing costs associated with manual decision-making processes - several critical gaps continue to exist in the literature. One gap exists concerning evaluation windows. Most studies evaluating LSTMs/CNN-LSTM models for forecasting stock prices during historically positive/bullish or mixed-regime market conditions offer these models an inherent competitive advantage due to the positive trend in equity markets. Conversely, when these same models are applied during bear markets/higher volatility regimes - their performance generally deteriorates dramatically. Few studies report how these architectures perform when the market is genuinely falling -when three or four equities in a portfolio decline by 15-30% during the evaluation period - although this represents a primary concern for managing risks/facilitating portfolio protection.

The second gap pertains to incorporating sentiment analysis into deep learning forecasting frameworks. Although extensive research has focused on studying NLP/Sentiment Analysis as a stand-alone predictor of market movements - little work has investigated combining sentiment with technical indicators inside a unified deep learning architecture. Two primary questions remain unanswered:

Does adding sentiment to a CNN-LSTM model that already receives 16 technical indicators enhance predictive accuracy; or does it simply introduce noise? Additionally, does Temporal Self-Attention - a mechanism enabling the model to selectively weight which timesteps in the lookback window are most informative - impact the answers to either question?

Utilizing technical data spanning January 2015 through May 2020 for training purposes and May 2020 through September 2021 for validation purposes, along with one year of sentiment-enriched data (September 2021 through September 2022) as the holdout test set - this project develops/trains a CNN-LSTM hybrid model with temporal self-attention across 84 unique model-ticker-condition configurations. Utilized as the evaluation window, September 2021 through September 2022 corresponds to the Federal Reserve's most aggressive rate hiking cycle since 1980. During this interval, three equities within this study declined by 15-30% while the CBOE Volatility Index averaged 24.11 - substantially exceeding its long-run average level of ~18. Therefore, this evaluation window represents one of the most difficult out-of-sample test environments presented within published research employing deep learning techniques for financial forecasting purposes - and therefore represents a realistic testing environment for demonstrating genuine predictive ability through accuracy metrics >50%.

In terms of practical importance - directional forecasting models possessing even a small yet consistent edge - i.e., correctly identifying direction three or four percentage points more than would be expected by random guessing - possess substantial economic value as they enable automated trading systems executing hundreds of trades per day. Moreover - understanding which features contribute to a model's predictions - and when in the look-back window a model is most attentive - provide actionable insights for practicing professionals: which information sources should be trusted for which equities; how long a look-back window should be utilized; and whether sentiment signals are reinforcing or contradictory with respect to specific equities.

Finally - the XAI component of this study - conducted via integrated gradients and attention-weight visualization - addresses this need.

1.4 Research Questions

This study addresses three specific research questions:

RQ1. Are technical indicators (EMA/RSI/MACD/Bollinger Bands/VIX) useful in enhancing next-day directional accuracy beyond raw OHLCV data; by how much; when assessed against a bear-market test window?

RQ2. Is incorporating daily sentiment scores derived from social media and/or financial news useful in enhancing predictive accuracy beyond what technical indicators provide; does the enhancement occur consistently across all ticker symbols?

RQ3. Does temporal self-attention between CNN Layers and LSTM Blocks add predictive capability; does its added predictive capability depend on composition of feature-set being utilized (technical-indicator only vs. technical + sentiment)?

1.5 Proposed Project Title

"A CNN-LSTM Hybrid Model for Multivariate Stock Price Direction Forecasting using Technical & Sentiment Indicators"

2. Securing Data

This portion of the report shows how each of the two sources of data needed for the project has been identified, located, and determined to be valid for the purpose of this research. In addition, there is a description of how all of the data for this project was secured prior to analysis.

2.1 Data Sources Identified and Accessed

There were two separate data source used in this project, each representing a unique time frame and set of features. Neither source is restricted in terms of accessibility.

2.1.1 Technical Price and Indicator Data — TradingView

Price and technical indicator data for all five stocks (AAPL, AMZN, GOOG, MSFT, TSLA) on a daily basis were downloaded from trading view. The files covered the period of 2 January 2015 through 30 October 2025 which equates to 2724 trading days per ticker across 17 columns. The first 5 price-volume columns represent raw market information while the remaining 11 columns contain pre-calculated technical indicators. This full dataset was limited to use up to september 2021 for model training and validation with the test window being defined by sentiment data that covers september 2021 through september 2022.

The technical indicators selected include the 20-day and 50-day exponential moving averages (EMA_20, EMA_50), the 14-day relative strength index (RSI) and its smoothed signal line (RSI_MA), the MACD oscillator and its signal line and histogram (MACD, Signal, Histogram), the Bollinger Band base, upper band, and lower band (BB_Basis, BB_Upper, BB_Lower), and the CBOE Volatility Index (VIX). These indicators were selected because they are among the most commonly utilized technical indicators in technical analysis and have also been used in similar deep learning-based forecasting studies (Sezer & Ozbayoglu, 2018; Fischer & Krauss, 2018; Lu et al., 2020). All five files were downloaded using excel in a format with no missing values in any of the technical columns.

2.1.2 Sentiment Score Data — Kaggle (Equinxx, 2023)

Pre-aggregated sentiment scores were collected from social media posts and financial news coverage for the same five tickers over a one year time frame from 30 September 2021 to 29 September 2022. The sentiment scores are contained within a Kaggle dataset titled "Stock Tweets for Sentiment Analysis and Prediction" created by Equinxx (2023). Since all five tickers use the exact same calendar and the sentiment scores are based upon aggregated social media activity for each ticker, this dataset provides a means to capture the market's perception of each company during this time frame. Additionally, since the data is openly accessible and can be downloaded directly from the Kaggle platform without requiring institutional access or requesting permission to access the data, this data meets both requirements.

Immediately following downloading the data a quality verification process was completed. The Google sentiment file was found to contain 14 missing sentiment values (from a total of 252 observation days). With regard to all ten files this is the only quality issue encountered. These 14 values were subsequently filled-in using forward fill (the previous day's observed sentiment value) when completing the Data Preparation phase. All other sentiment files were complete with zero missing values.

In addition to verifying data quality during Exploratory Data Analysis revealed another major data provenience problem. The correlation coefficient (Pearson r) of $r=1.00$ was calculated between the sentiment series for Amazon.com Inc. and Microsoft Corporation. This result demonstrates that both companies' sentiment series were generated from the same aggregation output by the same news feed pipeline. Thusly, neither company produced genuine sentiment signals specific to their respective business operations. Therefore, Microsoft Corporation was eliminated from consideration in Condition C3 during the modeling phase due to duplicate

information being present. However, this decision is documented as a limitation of this study in Section 6.

2.2 Dataset Summary

Ticker	Technical File (T)	Sentiment File (S)	T: Rows	T: Cols	S: Rows	S: Cols	Missing Values
AAPL	2015-01-02 to 2025-10-30	2021-09-30 to 2022-09-29	2,724	17	252	18	0
AMZN	2015-01-02 to 2025-10-30	2021-09-30 to 2022-09-29	2,724	17	252	18	0
GOOG	2015-01-02 to 2025-10-30	2021-09-30 to 2022-09-29	2,724	17	252	18	14 (Sentiment, imputed)
MSFT	2015-01-02 to 2025-10-30	2021-09-30 to 2022-09-29	2,724	17	252	18	0 (excluded from C3)
TSLA	2015-01-02 to 2025-10-30	2021-09-30 to 2022-09-29	2,724	17	252	18	0

Table 1. Complete summary of data secured for all five tickers across both sources

2.3 Data Access Confirmation

Prior to beginning the analytical work with both data sources were verified as accessible and complete. The TradingView data is available to users registered on the platform At an affordable subscription price; additionally, the Kaggle dataset is available under an open license and free to download. All ten files (five T-files and five S-files) were downloaded, verified for completion, and imported into the analytical environment. Verification was completed for calendar alignment issues regarding all five tickers having identical trading schedules with no date mismatch across the entire technical timeframe of 2724 days and sentiment timeframe of 252 days. This verifies that no interpolation or alignment adjustments were required before commencing model development.

Note on Data Cost and Quality: While there is high quality and ticker specific sentiment data available from professional sources (Bloomberg, Refinitiv, Financial NLP API Services) it usually has to be paid for with a subscription cost which would exceed the budget of this academic study.

3. Background and Review of the Literature

This section examines existing literature related to applying Deep Learning techniques to predict future directions of stock price based on Technical Indicators and Sentiment Indicators. Literature is grouped by topic - first, we examine theoretical bases of stock price prediction, including the Efficient Markets Hypothesis. Next, we examine prior work in the area of using classical machine learning techniques for stock price prediction. Following that, we evaluate prior work using Recurrent Neural Network (LSTM) architectures. We also examine prior work using Convolutional Neural Networks (CNN). After that, we look at prior work using Hybrid CNN-LSTM architectures. Finally, we conclude with an examination of prior work using Attention Mechanisms and Explainability Methods. Thirteen works from various fields (Finance, Information Systems, Machine Learning, Data Mining) are examined.

3.1 Theoretical Foundations: Market Efficiency and Its Limits

Market price prediction begins with the Efficient Market Hypothesis (EMH) proposed by Fama (1970) in "The Journal of Finance." The EMH posits that at all times asset prices reflect all known information regarding those assets. Therefore, there is no possible way to systematically forecast market prices. As stated above, the strong form of the EMH further states that no investment strategy - whether technical, fundamental or data-driven -

will be able to consistently beat a passive market index after adjusting for risk and transaction costs. Decades of empirical studies, however, have found many instances where EMH appears to fail. There exist several documented examples of predictable pricing anomalies in financial markets. Examples include momentum effects, short term reversals, earning surprise drift and sentiment-driven mispricing. These anomalies occur in virtually all major world-wide equity markets and are strongest in individual securities at daily time scales. It does not appear that EMH completely fails; instead, it suggests that even sophisticated investors may take time to process large amounts of new information, that behavioral errors provide systematic deviations from rational equilibrium prices and that powerful algorithms running quickly may be able to find exploitable patterns before such patterns are fully arbitrage free.

In the study presented here, we demonstrate how challenging this problem is empirically. None of our 16 technical indicator measures had a highest absolute linear correlation greater than .045 with the next day's price movement for any of the five tech companies studied. Most technical indicator measures were correlated with less than .016. Since all of our technical measure correlations with price direction were so low, we show that any possible predictive signal would likely be non-linearly contained in some temporal relationship among the measures rather than any particular measure itself. Therefore, we justify the use of non-linear deep learning models over traditional linear classification methods. Our empirical demonstration also serves as the data-driven basis for our modeling assumptions made in this study.

3.2 Classical Machine Learning for Financial Prediction

Machine learning models were previously widely adopted in finance before deep learning became popular. Tree-based ensemble models were shown to be effective for predicting the directionality of individual stocks using technical indicators. Specifically, Khaidem et al., (2016) demonstrated that Random Forest classifiers achieved better-than-chance directional accuracy using technical indicators for individual stocks. Thus, ensemble tree-based models established themselves as a non-linear classical baseline for comparing against deeper models. However, as noted previously, tree ensembles are limited to capturing cross-sectionally associated relationships between input features and next day's direction. They do not capture the temporal order of input observations.

Another classical baseline model that has been recently developed is the XGBoost algorithm developed by Chen and Guestrin (2016) at the ACM SIGKDD conference. XGBoost has rapidly become very popular in applied machine learning due to its regularization properties, computational efficiency and good performance on tabular data. Across a wide range of financial prediction tasks XGBoost has often performed equally as well as Random Forest models. Both models are included in the current study as classical benchmarks. What they lack is the ability to handle sequential data since they both treat the 20-day input window as a flat unordered set of feature vectors - they lose all information about the temporal position of each observation. The high negative correlation between Random Forest feature importances and CNN-LSTM integrated gradients attributions of $r = -0.74$ demonstrates that these two methods derive fundamentally different types of insights from the same data. Hence, the need for sequential deep learning is validated.

3.3 LSTM Networks for Financial Time-Series Prediction

Long Short Term Memory networks were developed by Hochreiter and Schmidhuber (1997) in Neural Computation. The introduction of Long Short Term Memory networks solved the vanishing gradient problem that prevented earlier recurrent network models from being effectively used for long sequences by allowing three learned gates - input gate, forget gate and output gate - to control the flow of information through a persistent cell state. With this mechanism Long Short Term Memory models can remember valuable information from arbitrarily far back in a sequence and therefore are ideal for analyzing and predicting behavior

of financial time series where important events may last for days or weeks.

Fischer and Krauss (2018), who published in European Journal of Operational Research, provided what was ultimately the most influential application-oriented study linking Long Short Term Memory models to financial predictions. Using only historical price return values as inputs to their LSTM models on each constituent company of the S&P 500 Index they reported average directional accuracy levels of **53.2%**. Their results represent a 3.2 percent directional accuracy advantage over pure random guessing. We use Fischer and Krauss' (2018) 53.2% benchmark as our primary external comparison for directional accuracy since our CNN-LSTM Attn model achieves it under Conditions C3, although we evaluate on a much harder bear market window than Fischer and Krauss did.

Nelson et al., (2017), who published in Proceedings of IEEE International Joint Conference on Neural Networks, reported directional accuracy rates ranging from 55% to 56% using individual stock data and Long Short Term Memory models. However, Nelson et al.'s results utilized an unrealistically optimistic evaluation regimen with respect to time scale and a randomly assigned training/evaluation split. As a consequence their accuracy rates are inflated artificially and do not accurately estimate real-world performance potential for these models.

We make adjustments to their methodology by utilizing walk forward cross-validation with a four fold expansion and a bear market window identified through central bank monetary policy actions for our evaluations of this study's results. While our accuracy estimates are consequently somewhat pessimistic relative to those of Nelson et al., they are more representative of real-world deployable performance potential than Nelson et al.'s overly optimistic estimates.

3.4 Convolutional Neural Networks for Temporal Pattern Extraction

Sezer and Ozbayoglu (2018), who published in Applied Soft Computing Journal demonstrated that adapting one dimensional Convolutional Neural Networks (CNN) for financial time series forecasting resulted in superior performance over other architectures for identifying local patterns over time - momentum signals, reversal patterns, etc.. Sezer and Ozbayoglu showed that CNN's are capable of detecting patterns within short windows of price data with significantly reduced computational requirements compared to Recurrent Neural Network (RNN) architectures like Long Short Term Memory (LSTM). This is because CNN's are optimized for detecting locally structured repeated features within a sequence.

A limitation of solely relying on CNN's for identifying patterns over time is the loss of temporal ordering when global pooling is applied. When global pooling is applied to reduce a variable length input window down to a fixed length representation all timing information is lost - i.e., when a particular pattern occurred in the sequence. We confirm this limitation empirically via our CNN-only ablation model which performs poorly across all three feature condition configurations versus our full CNN-LSTM hybrid model. Therefore, detecting local patterns without considering sequential memory is inadequate for predicting future directions in next day's stock price movements.

3.5 CNN-LSTM Hybrid Architectures

Lu et al., (2020) published in Complexity Journal introduced the CNN-LSTM hybrid architecture that forms the basis of our study's main framework. The authors propose applying one dimensional convolutional neural networks for extracting features over short ranges followed by passing the compacted representations through a LSTM layer for incorporating long range sequential memories. Results on multi-security datasets using technical indicators indicated that their CNN-LSTM hybrid architecture achieved 55.9% directional accuracy while outperforming both CNN-only and LSTM-only ablated versions in the same experiment design. This paper serves as our primary source of justification for designing our hybrid CNN-LSTM ablation structure.

It is worth noting that the difference between Lu et al.'s 55.9% and our 51.3% results under identical conditions (C2) are almost entirely attributable to differences in evaluation regimes. Lu et al. evaluated during periods characterized by mostly positive market conditions whereas our study evaluated September 2021 through September 2022 - a period during which three of the five technology stocks dropped by 15-30% and marked the beginning of a clearly defined structural regime shift caused by a tightening cycle of interest rate increases by central banks. A model trained on several years of predominantly bullish market conditions and evaluated on a bear-market period constitutes an out-of-distribution challenge that increases the difficulty level of our task relative to Lu et al.'s task conditions.

3.6 Sentiment Analysis in Financial Markets

Bollen, Mao, and Zeng (2011) demonstrated that aggregate daily mood scores obtained from Twitter posts via sentiment analysis can forecast future movements in the Dow Jones Industrial Average with accuracy rates higher than 80%. Although this rate has not been reliably duplicated by other researchers at comparable levels in subsequent work, Bollen et al. (2011) demonstrated that publicly-expressed sentiment conveys predictive information about market trends beyond what signals derived solely from prices convey. Several follow-on studies have since validated this conclusion with regard to various types of data sources, markets, and time frames. Bollen et al. (2011) established the intellectual basis for sentiment-enhanced financial prediction and prompted the incorporation of the Sentiment feature in Condition C3 of this study.

This study demonstrates that the directional influence of sentiment varies by ticker and by regime. Positive sentiment is associated with upward price movement for Apple. However, for Google and Amazon, sentiment is contrary in the 2022 bear market: positive news was associated with downward price movement; and, consistent with a risk-off macroeconomic environment, favorable company-specific news was outweighed by adverse interest-rate impacts. For Tesla, sentiment did not appear to possess much predictive value due to its nearly constant distribution (standard deviation = .057) throughout the test period. Therefore, these varying results demonstrate that it is not valid to assume that sentiment will always enhance the predictive power of any stock or any given market regime.

3.7 Attention Mechanisms for Sequential Neural Networks

Bahdanau, Cho, and Bengio (2015) presented the additive attention mechanism in the context of neural machine translation. Instead of aggregating all past information into one fixed-size vector, attention allows the model to calculate a weighted average of all previously computed vectors such that the weights are determined by the relevance of each previously-computed vector to the current prediction. In doing so, the model may selectively concentrate on those portions of the history that are most relevant to making a particular prediction, instead of considering all parts of the sequence equally.

Qin, Song, Chen, Cheng, Jiang, and Cottrell (2017) presented an adaptation of dual-stage attention for financial time-series predictions at IJCAI. Specifically, they implemented DARNN (dual-stage recurrent neural network) which utilizes attention both to select relevant input features at each time-step and to identify the time-steps themselves that are most informative across the entire sequence. As such, DARNN outperformed standard LSTMs on several multivariate time-series forecasting tasks. He, Zhang, Ren, and Sun (2016), at CVPR, described residual connections that enable the addition of an attention layer to existing architectures without affecting their ability to represent the input correctly. A portion of the original signal is passed through a "skip" connection and combined with the transformed signal after attention processing to ensure that no degradation occurs relative to simply passing the signal through without applying the attention. In terms of empirically validating these theoretical developments and establishing them as useful tools for financial modeling, this study makes a novel contribution to the extant literature: the benefits of temporal attention depend critically upon the composition of the feature set. Adding attention to CNN-LSTM with only technical inputs (Condition

C2) resulted in a reduction in mean accuracy of 2.1 percentage points. Conversely, adding the identical attention mechanism to CNN-LSTM with sentiment inputs (Condition C3) increased mean accuracy by 2.4 percentage points. Hence, the conditional effects observed here indicate that attention is valuable only when there are qualitatively disparate and/or sparse signals contained in the feature set; otherwise, attention adds unnecessary noise.

3.8 Explainability and Attribution Methods for Deep Learning

Lundberg and Lee (2017) developed SHAP (SHapley Additive exPlanations), a universal model interpretability framework grounded in cooperative game theory. SHAP assigns each input feature a SHAP value indicating its incremental contribution to the output value averaged across all possible combinations of feature subsets. SHAP satisfies key desirable mathematical properties including efficiency (the total of all feature contributions equals the predicted value), symmetry, and monotonicity. SHAP has become very popular in many high-stakes applications like finance, medicine, and credit-risk assessment for explaining black-boxes.

Sundararajan, Taly, and Yan (2017), at ICML, proposed Integrated Gradients as another axiomatic gradient-based attribution method. Integrated Gradients integrates the model's gradient with respect to each input feature along a path from some baseline (e.g., zero) to the actual input generating an attribution value that satisfies both the sensitivity axiom and the implementation-invariant axiom. Like SHAP Gradient Explainer, Integrated Gradients is mathematically equivalent for attributing feature importances but does not need modifications to the model's gradient computation system; hence it was crucially advantageous for this study to implement Integrated Gradients purely with TensorFlow since importing SHAP corrupted TF's gradient registry for good in the same session.

Integrated Gradients reveal that CNN-LSTM rely almost exclusively on the two most recent days in the 20-day lookback window ($t-2$ and $t-1$ have by far the largest attribution values), thus providing strong evidence for a short-horizon momentum-detection mechanism. This result is significant for practitioner use: the model appears well-suited for implementing overnight holding strategies and reacting mostly to recent price-movements rather than longer term historical influences.

3.9 Machine Learning in Empirical Asset Pricing

Gu, Kelly, and Xiu (2020), publishing in *The Review of Financial Studies* provided an exhaustive comparison of machine-learning techniques used in predicting equity return using 94 firm-characteristics across a massive panel of U.S. equities. Gu et al. (2020) document that neural networks out-perform linear factor models and tree-based approaches especially when the number of firm-features is large and relationships among features are nonlinear. Additionally Gu et al. (2020) find that transforming trading-volume logarithmically improves neural-network performance by dampening extreme-trading event influences on neural-networks' outputs. This result provided direct motivation for log1p volume-transformation performed in this study and exemplifies how results from large scale empirical asset pricing research can inform practical preprocessing-decisions in deep-learning applied-studies.

3.10 Statistical Evaluation of Forecast Accuracy

Diebold and Mariano (1995) proposed a formal statistical test for evaluating whether one predictive model generates superior forecasts compared to another. Diebold and Mariano's test computes a time-series of losses generated by subtracting forecasts produced by each model from true outcomes and tests if the mean-loss differential is statistically differentiable from zero using heteroskedasticity-autocorrelation-consistent-variance estimation. Harvey, Leybourne, and Newbold (1997) proposed a small-sample adjustment to Diebold-Mariano test designed to prevent excessive rejection of null hypotheses when test-period length is limited. Since this study utilized a test-window containing only 231 observations, application of this adjustment was necessary; it

provides approximately 20–30% power to detect 2 percentage-point accuracy-differences between competing models; therefore most real accuracy-differences among models would fail to achieve traditional statistical significance. Thus, failure to obtain wide-spread statistical significance in 70 DM tests conducted in this study should be interpreted as expected consequences of sample-size limitations rather than evidence that the models are functionally equivalent.

3.11 Summary of Established Literatur

Financial markets do not exhibit perfect efficiency at short horizons and therefore non-linear models can discover patterns in individual equity daily returns; as documented in prior literature regarding momentum anomaly and sentiment anomaly; as shown by non-zero predictive accuracy achieved by deep learning models in multiple independent studies.

Deep learning models out-perform classical machine learning for financial time-series prediction because sequential structure contains predictive information inaccessible to order-agnostic models; as shown by prior literature (Fischer & Krauss, 2018; Lu et al., 2020); as supported by RF vs CNN-LSTM Spearman $r = -0.74$ relationship observed in this study.

Hybrid CNN-LSTM models clearly out-perform both ablated versions; thereby confirming prior findings that combining local-pattern extraction capabilities (CNN) and long-term sequential memory functions (LSTM) enhances performance on financial-time series-prediction-tasks (Lu et al., 2020). The findings in this study replicate this result across three feature conditions and five tickers.

Sentiment possesses directional-information beyond price-based signals; although magnitude, directionality and consistency of this effect vary significantly depending on ticker-symbol or economic-regime; as demonstrated by prior research regarding sentiment analysis in financial markets (Bollen et al., 2011).

Attention mechanisms are selectively-beneficial-not-universally-so: this study presents new evidence that the degree to which temporal attention benefits performance depends heavily upon whether feature-set consists of heterogeneous-sparse-signals versus correlated-dense-technical-indicators.

Evaluation Regime Is Critical But Under Reported Variable: Benchmark accuracy figures from prior research typically derive from bull-market evaluations whereas this study evaluated models utilizing a bear-market evaluation window representing a significantly-harder-test-of-model-generalizability. These established findings define the methodological choices made in the present project and frame its original contributions within the cumulative knowledge of the field.

4. Methodology

In this part I will describe the entire methodology used in this study for collecting the data, the processes of cleaning and preparing the data and the methods of analysis to solve the three research questions mentioned in Section 1. Following the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology structure, this study is divided into five sequentially linked phases: Data Understanding, Data Preparation, Modelling, Evaluation and Explainability. Below, each one will be detailed.

4a. Type of Data Used

There have been several studies based on financial data, but there are few studies based on technological stocks. In this case, two different types of quantitative time-series data have been used, both measured at a daily frequency. For this purpose, the five large-cap U.S. Technology Stocks: AAPL (Apple), AMZN (Amazon), GOOG (Alphabet), MSFT (Microsoft), and TSLA (Tesla) have been selected as well as their respective technical-price and indicators data.

Technical Price and Indicators Data

The primary database includes daily time-series data for each ticker for the time span 2nd January 2015 to 30th October 2025. Therefore, for each ticker, 2724 trading days and 17 columns of data exist. Of this full file, only data up to September 2021 was used for model training and validation, with the test window defined by the sentiment data covering September 2021 to September 2022. The five basic price-volume fields -Open, High, Low, Close and Volume - correspond to the raw market data generated every trading day. The 11 technical indicators derived from these fields - EMA_20, EMA_50, RSI, RSI_MA, MACD, Signal, Histogram, BB_Basis, BB_Upper, BB_Lower and VIX -are calculated by TradingView through standard financial equations. These indicators are designed to capture different facets of price movement: trend-direction (Bollinger Bands and EMAs); momentum (RSI and MACD); and overall-market volatility (VIX). The data are organized (structured), numerical and ordered temporally with no randomization nor cross-sectional mix-up throughout the analysis process.

One important property of this data is its very high degree of non-stationarity. ADF-tests carried out during the Data Understanding Phase confirmed that raw price-levels (Close, EMA_20, EMA_50 and Bollinger-Bands) are non-stationary at all conventional significance levels (all ADF-pvalues >0.99), i.e., their statistical properties vary over time. By contrast, all three momentum oscillators (RSI, MACD and Histogram) are stationary. This distinction motivated the definition of the prediction task: instead of trying to forecast future price-levels - a task that requires modeling non-stationarities - the objective of this project is to predict whether tomorrow's price will move up or down (binary-classification-task), thus forecasting a stationarity-by-construction quantity.

Sentiment Score Data

The second database provides a pre-aggregated daily sentiment-score for each ticker between 30th September 2021 and 29th September 2022. Therefore, for each ticker, there are 252 trading-days rows and 18 columns (i.e., the same 17 technical columns as the primary-database, plus a Sentiment column). The sentiment-score corresponds to a numeric-value derived from social-media-posts and financial-news-coverage published on each trading-day. A positive number indicates net-positive sentiment; a negative number indicates net-negative sentiment. Like the Technical-Dataset, the Sentiment-Dataset is structured, numerical and temporally ordered. Two documented-quality-problems were recognized in the Sentiment Dataset during the Data Understanding Phase. Both problems were treated during the Data Preprocessing Phase. Firstly, Google's Sentiment Series had 14 missing values (i.e., 5.6% of all observations) that were replaced by forward-filling. Secondly, it was observed that the Sentiment Series of AMZN and MSFT were identical to each other with Pearson $r=1.000$. It was therefore concluded that there existed a Provenance Problem in the original Sentiment Dataset. Henceforth, Microsoft was dropped from all analyses involving Sentiment Condition-C3.

Conditions for Experiments

To evaluate systematically how much each Information Source contributes to the performance of predictive models independently from the contributions made by each of the other Information Sources - which may interact positively or negatively with each other - three feature-conditions were defined to compare the results in a controlled way:

Condition	Name	Features Included	No. Features	Tickers
C1	Raw OHLCV	Open, High, Low, Close, Volume	5	AAPL, AMZN, GOOG, MSFT, TSLA
C2	Technical	C1 + EMA_20, EMA_50, RSI, RSI_MA, Histogram, BB_Basis, BB_Upper, BB_Lower, MACD, Signal, VIX	16	AAPL, AMZN, GOOG, MSFT, TSLA
C3	Sentiment	C1+C2 + Sentiment	17	AAPL, AMZN, GOOG, TSLA

Table 2. Experimental conditions defining the three feature sets used across all models.

4b. Methods of Data Collection and Sources

Both datasets can be retrieved from public-accessible platforms without any institutional-restrictions regarding access to them.

Technical Data — TradingView

Data consisting of technical price and indicator data for all five tickers were downloaded from TradingView (<https://www.tradingview.com>). TradingView is one of the largest financial charting platforms with registered users. They provide historical complete price data since each stock began trading. Also, TradingView has built-in technical indicator calculations using standard formulas from quantitative finance. Data was exported from Trading View in Excel-format using the Export button within the application. All five files were pulled at the same time so as to ensure consistency with respect to when the data was pulled. There were no missing values in any technical field across all five tickers.

All technical indicators included were chosen according to three criteria: (1) they are some of the most popular technical indicators used in empirical studies about Quantitative Finance as shown in our Literature Review;(2) they address fundamentally different aspects of price movement - trends (Bollinger Bands & EMAs); momentum (RSI & MACD); and volatility (VIX) - thus maximizing feature diversity without including redundant features; and,(3) they are pre-calculated by default through TradingView without having to write custom scripts.

The VIX was added as a measure of overall-market-volatility to create an additional context-variable beyond those obtainable solely from individual stock-prices.

Sentiment Data — Kaggle (Equinxx, 2023):

The Daily Sentiment Scores were obtained from the Kaggle dataset called "Stock Tweets for Sentiment Analysis and Prediction" by Equinxx (2023), available at <https://www.kaggle.com/datasets/equinxx/stock-tweets-for-sentiment-analysis-and-prediction>. This dataset is available for free under an open data license and does not include any download restrictions. The Five Tickers' Specific Sentiment Files were downloaded in CSV Format directly into Python's Analysis Environment using Pandas. The data covered Septemeber 2021- Septemeber 2022 - matching the test-period defined for this Project.

4c. Data Processing and Cleaning

As the original 7-step data processing pipeline was designed to remove all types of data leaks when transforming downloaded files into model ready arrays and since every single type of transformation applied to the data is first to training data then to the same data with respect to validation and testing, there will be no way to use future data to make a decision about how to process historical data.

Step 1 — Sentiment Imputation

Google's sentiment file had 14 missing values (NaN) in the sentiment column. These were filled using a forward fill method where each NaN was replaced with the most recent non-NaN value. The reason for doing it this way is for causal validity: only information up to and including time t can be used to determine what happens at time t . Using future information (sentiment) would violate causality. The interpretation is that when no tweet or news articles are available, we assume that the previous day's sentiment is representative of the overall sentiment of the market for the current day. In addition to forward filling missing values, a backward fill was performed as an additional check. As a result of both methods, there are no missing values left in the entire dataset across all 10 files.

Step 2 — Target Variable Construction

The binary target variable $y[t]$ was created based upon whether the closing price ($\text{close}[t]$) of the stock was greater than the closing price ($\text{Close}[t+1]$) for the next trading day. If $\text{Close}[t+1] > \text{close}[t]$ then $y[t] = 1$ (up) and if $\text{Close}[t+1] \leq \text{close}[t]$ then $y[t] = 0$ (down or flat). The targets were made from the raw close price column prior to normalization or other transformations so that the direction indicator does not depend on any choice made for normalization. To avoid having a look ahead bias, the $\text{Close}[t+1]$ price that is used for labeling was never used as an input feature in any model. The last row of each file has been removed as there is no $t+1$ close price for the last trading day. The mean proportion of up days in the training data ranged from 50.2% (AAPL) to 53.0% (TSLA) which indicates a slight bullish bias over the 2015-2020 training period consistent with a bull market environment. Class weights were calculated using the training distribution and were used throughout training for all deep-learning models.

Step 3 — Temporal Split

The data was divided into training, validation, and test sets using a fixed time-split without any shuffling at any point. The splits are listed below:

Split	Date Range	Source File	Rows	Sequences
Training	2015-01-02 to 2020-05-27	TradingView (T) file	1,359	1,339
Validation	2020-05-28 to 2021-09-29	TradingView (T) file	339	319
Test	2021-09-30 to 2022-09-28	Kaggle Sentiment (S) file	251	231

Table 3. Temporal split boundaries. Sequence count = rows minus the 20-day look-back warm-up period.

An enforcement statement for leakage was added to the code: the largest date in the training set must be strictly smaller than the smallest date in the test set. This condition was confirmed to be true for all five stocks before any scalars were fit. There was no shuffling involved at any point, as shuffling financial time-series data violates the assumption of causality and introduces look-ahead-bias, allowing future knowledge of the series to affect the probability distributions of the training set.

Step 4 — Volume Transformation

Before scaling, all Volume values were logarithmically transformed via $\log_{1p}(x) = \log(1+x)$. This was done because raw trading volumes have skewed distributions with skewness measures ranging between 2.0 and 2.9 for all five stocks. Extreme values (e.g., during earnings announcements, etc.) could overwhelm the LSTM's input-gate activation functions and disrupt gradient flows. The distribution of volumes became nearly symmetric after applying the log-transformation, reducing skewness to approximately between 0.24 and 0.63

across all five stocks. We drew motivation for this preprocessing step from Gu, Kelly & Xiu (2020), who demonstrated substantial improvements in performance from applying a log-Volume transformation in their large-scale asset-pricing models.

Step 5 — Feature Scaling

Standardization was applied to all features using StandardScaler (mean=0, std=1) - fitted exclusively on the training data - and applied uniformly to both validation and test data. It is crucial to fit the scaler only on the training data, as fitting on all data would provide information regarding future distributions of feature-values to the scaler parameters, thereby increasing estimated test-performance metrics.

Due to the nature of the sentiment-data - zero-filled placeholders (for no sentiment data existing for the 2015-2021 training period), Condition C3 necessitated a two-stage scaling methodology. When creating a scaler during training, the sentiment column contained only zeros (as there was no actual sentiment-data available for those periods). Therefore, fitting a scaler on those zeros resulted in a scale-parameter equaling zero, effectively rendering the sentiment feature useless at test-time. To rectify this issue, a second scaler was trained on actual sentiment data from test-period (s)-file - and its statistics (mean/std) were subsequently appended into a common scaler trained on all features combined (17 total features). This allows us to apply meaningful z-score ranges to the sentiment feature at test-time — while maintaining zero in training period as a neutral prior.

Step 6 — Sliding Window Sequence Construction

All three feature-conditions were transitioned from 2D tables (rows * columns) to 3d sequence-tensors (samples * steps * columns) utilizing a sliding window of size twenty-trading-days in a causal manner. For example, given position i in our processed array of scaled features — the sequence-input $X[i]$ includes rows i thru $i+19$ (dimensions: $20 \times \text{columns}$), whereas label $y[i]$ denotes the direction-indicator at position $i+20$. Note that this creates a fully-causal relationship: the label at position $i+20$ is never observable within $X[i]$. We selected a window-size of twenty-trading-days to align with approximately one-month of trading history. Our selection aligns well with previously-established window-lengths utilized by Fischer and Krauss (2018) and lu et al. (2020).

Step 7 — Verification Checks

After completion of all preparatory-pipeline operations — we implemented an extensive automated verification suite — which confirmed: (1) that there were no NaN values present in any of six arrays ($x_train/X_val/x_test/y_train/y_val/y_test$) for any combination of stocks/condition; (2) correct dimensions for all sequence-arrays — i.e., ($n_samples, 20, n_columns$) for x-arrays and ($n_samples,$) for y-arrays; (3) binary indicators alone in all y-arrays — e.g., either 0 or 1; (4) non-zero scales produced by scaler-fits; (5) non-zero variances in all test-array features. In summary, all fourteen combinations of stocks/condition passed all tests.

4d. Analysis methods

In the analysis stage of this research, we have created and run six different models on Three different feature conditions. Thus, in total, we have analyzed 84 fully-trained models. The entire study has been done in Python 3.0 on google colaboratory pro using TensorFlow 2.19.0, Keras 3.13.2, scikit-learn and XGBoost. In order to reproduce our results, we have used a fixed random seed (seed=42) in all stochastic parts of our code (python, numpy, TensorFlow).

Classical machine learning methods (baselines)

We have trained Three classical machine learning baselines on the flat version of the sequence inputs ($n_samples \times (20 * n_features)$). We treated the temporal sequence as a flat vector of feature values.

logistic regression ($c=0.1$, L2 regularization, max iterations = 2000, class weights = 'balanced'): a linear model with L2 regularization as a lower bound of what can be achieved by these models.

near-chance accuracies of the logistic regression and other two models confirm the findings from the data exploration part that there is no relevant linear predictive information in the data ($\max |\text{corr}| = 0.045$).

Random Forest (200 trees, max_depth = 8, class_weights = 'balanced'): an ensemble of decision trees that learn nonlinear relationships between individual feature values but treat the sequence as an unordered table (i.e. Loses temporal structure).

XGBoost (200 estimators, max_depth = 4, learning_rate = 0.05, scale_pos_weight = training class ratio): a strong nonlinear ensemble model that uses boosting to improve predictions based on earlier weak models. All Three classical models were trained on the union of training and validation sets (1698 samples) in order to maximize the number of training samples available for testing.

Deep learning methods

We also developed Three deep learning architectures that preserve the temporal structure of the 20 day window. These models use 3d sequence inputs ($n \text{ samples} \times 20 \times n_{\text{features}}$):

CNN-only (ablation): extracts local temporal patterns in the first 20 days of a sequence using two Conv1D layers with causal padding. It then collapses the time dimension by averaging over all timesteps using global average pooling. Isolating the contribution of local feature extraction without sequential memory.

LSTM-only (ablation): feeds the 20-day sequences into a single LSTM layer (64 units) directly without any preceding convolutional processing. Isolating the contribution of sequential memory without local pattern extraction.

CNN-LSTM (proposed hybrid): first, it extracts local temporal patterns using a conv1d layer with causal padding and 32 filters, kernel size 3 and ReLU activation. Then it halves the temporal dimension by applying MaxPooling1D with pool_size=2. Next, it applies Dropout(0.3) to prevent co-adaptation. Afterward, it feeds the compressed representation into an LSTM layer (64 units) that integrates sequential dependencies over the compressed representation. Finally, it applies a second Dropout(0.3), a Dense layer (32 units with ReLU activation) before finally computing the Sigmoid output.

CNN-LSTM-attn (proposed hybrid with attention): is identical to CNN-LSTM with a single-head temporal self-attention layer (bahdanau et al., 2015) placed between CNN block and LSTM. Computes a scalar weight for each of the ten compressed timesteps as follows:

+ $\text{score}(t) = \tanh(w \cdot h_t + b)$,
+ $\alpha(t) = \text{softmax}(v \cdot \text{score}(t))$,
+ $\text{output} = \alpha(t) \cdot h_t$.

A residual connection (he et al., 2016) is used to add the original CNN output back to the attention output to stabilize training while preventing the attention weights from being degenerate. The attention mechanism adds only 1,152 parameters, increasing total parameters from 28,769 to 29,921 - less than 4% increase.

All deep-learning architectures use the same regularizer: L2-weight-decay ($\lambda = 10^{-4}$) on all weight

matrices and Dropout(30%) at two locations within networks and batch normalization after LSTM layers. We used Adam optimizer with initial learning-rate = 0.0001 and gradient-clipping(clipnorm = 1.0) as our optimization algorithm. We used binary-cross-entropy as our loss-function with class weights equalized during training.

Walk-forward cross validation training protocol

To avoid both early-stopping-bias (stops too early) and over-fitting-bias (trains too long), we used a four fold expanding walk-forward cross-validation protocol to find the optimal training depth for each deep-learning architecture. For each fold we added one additional year to training-data and tested on the next year. Best epoch selected through early-stopping was saved for folds 1, 2 and 3; fold 4 (the covid crash in 2020) was not considered in selecting the best epoch due to documented structural break in financial markets that is not representative of our test environment for interest rate hike in 2022. Final training-depth was defined as mean-best epoch for Three included folds rounded up to nearest integer. Following selection of training-depth, each final-model was re-trained on all pre-test data (training + validation) for exact-number-of-epochs corresponding to selected training-depth, without early-stopping.

Metrics for evaluation

For each model we evaluated four different metrics on holdout test-set:

Directional accuracy (primary metric)

The proportion of correct directional-predictions out of 231 test-day predictions where the prediction-direction (up/down) corresponds to actual next-day direction. This is most suitable for the binary directional forecast-problem and serves as base for all benchmark-comparisons.

ROC-auc

Measures how well model ranks up-days above down-days independently of classification-thresholds, ranges from 0.5(random) to 1.0(perfect).

Weighted-f1-score

Accounts for class-imbalance and provides balanced view of model's capability to correctly classify both up and down days.

Diebold-Mariano Test

Formally tests whether predictive-performance of two forecasting-models differs significantly under zero-one-loss using harvey et al. (1997)'s small-sample-correction. Rejects null if true predictive-performance-difference exists between models.

Integrated gradient attribution methodology for explainability

Model predictions were attributed using Integrated Gradients (sundararajan et al., 2017), which was implemented as a pure-tf computation using tf.GradientTape. We compute attributions for each of the 231 test-day predictions over 50-interpolation-steps from zero-input to actual-inputs, creating a 3-dimensional attribution-array (231x20xn_features). The average absolute attribution-per-feature collapsed-over predictions-and-timesteps defines overall feature-importance; average absolute attribution-per-timestep collapsed-over predictions-and-features defines temporal-importance. We calculate Pearson correlation between the learned-attribution-profile and the temporal-attribution-profile as cross-method-validation-checkpoint. For comparison, we trained rf on the same dataset as CNN-LSTM and compared impurity-based feature-importances to CNN-LSTM integrated-gradients feature-rankings using Spearman-rank-correlation.

5. Results

All data analytical techniques applied in this study are presented here. The presentation order is the same as the sequence of analyses conducted: first the iterative model training runs that resolved training issues, then the comparative accuracy results across all six architectures and three feature conditions, next statistical significance testing and finally, the explainable ai attribution analyses to reveal how and why the best performing models make their predictions.

5.1 training protocol development

The modeling phase of this project included four separate iterations of training; each iteration diagnosed an issue with the prior training protocol and implemented a single corrective action. This development process is important background information for understanding the final results of the project because it demonstrates that the choices made regarding the training protocols are responsible for a greater proportion of the accuracy improvements than the individual choices made regarding the architecture.

Training Run 0 (baseline) used a learning rate of .001 and included 128 LSTMs (total parameters = 106,625); most models stopped training between epochs 1-5 for nearly every ticker condition combination resulting in a mean CNN-LSTM accuracy of 47.1% on C1 and 47.2% on C2 - less than the random baseline of 50%. The AMZN C1 CNN-LSTM scored only 39.4%, which was significantly worse than all baselines. This failure occurred due to an overly large learning rate for financial data causing losses to spike immediately, at which point early stopping fired prior to any significant learning.

Training Run 1 reduced the learning rate to .0001, the number of LSTMs to 64 units and the number of CNN filters to 32 (total parameters = 28,769 with a parameter-to-sample ratio of 19:1 compared to 79:1 in training Run 0). Training extended out past epochs 6–40. The mean CNN-LSTM accuracy improved to 50.1% on C1 and 51.0% on C2 — a gain of 3.8 percentage points from training Run 0 to training Run 1 achieved solely through hyperparameter correction without changing the architecture.

Training Run 2 replaced the fixed validation split with four-fold expanding walk-forward cross-validation while adding gradient clipping (clipnorm = 1.0). The COVID crash window (fold 4) prompted early stopping to fire at epoch 1 for almost all models. This pulled the mean optimal epoch below 6 limiting training depth.

Training Run 3 excluded Fold 4 (COVID window) from the epoch vote while retaining it for training. This increased optimal training depth to 9–14 epochs for many models and raised mean CNN-LSTM accuracy to 51.1% on C1 and 51.3% on C2. CNN-LSTM became the best deep learning model overall in training Run 3 outperforming all classical baselines under all conditions.

Run	Key Change	CNN-LSTM C1	CNN-LSTM C2	CNN-LSTM C3	Net Change C2
Run 0 — Baseline	lr=0.001, lstm=128	47.10%	47.18%	50.76%	—
Run 1 — Hyperparameters	lr=0.0001, lstm=64	50.13%	51.00%	51.52%	+3.82pp
Run 2 — Walk-forward	4-fold CV + clipnorm	50.48%	50.82%	50.22%	−0.18pp
Run 3 — COVID excl.	Fold 4 from epoch vote	51.08%	51.25%	50.98%	+0.43pp
Run 4 — Attention	Attn layer added (C3)	50.65%	49.18%	53.35%	+2.37pp on C3

Table 4. Four-run training progression: mean CNN-LSTM directional accuracy across tickers per condition. Bold = best result.

5.2 The best models of directionality by conditions

The data reported in Table 3 is related to the directional accuracy, roc-auc and the weighted f1-score of the best model in each condition, for each stock ticker. The results of the conditions C1 and C2 are based on Run 3 (cnn-lstm) while the results of the condition c3 are based on Run 4 (cnn-lstm-attn). Each of these models has been tested using a total of 231 test sequences generated from the period that went from September 2021 to September 2022 which were identified as a bear market.

Cond	Ticker	Best Model	Accuracy	ROC-AUC	F1	vs 50% baseline
C1	AAPL	CNN-LSTM (R3)	50.65%	0.439	0.341	+0.65pp
C1	AMZN	CNN-LSTM (R3)	52.38%	0.554	0.360	+2.38pp
C1	GOOG	CNN-LSTM (R3)	47.19%	0.523	0.303	−2.81pp
C1	MSFT	CNN-LSTM (R3)	54.55%	0.514	0.405	+4.55pp
C1	TSLA	CNN-LSTM (R3)	50.65%	0.406	0.351	+0.65pp
C2	AAPL	CNN-LSTM (R3)	54.11%	0.568	0.519	+4.11pp
C2	AMZN	CNN-LSTM (R3)	52.81%	0.557	0.528	+2.81pp
C2	GOOG	CNN-LSTM (R3)	46.75%	0.529	0.369	−3.25pp
C2	MSFT	CNN-LSTM (R3)	51.52%	0.522	0.511	+1.52pp
C2	TSLA	CNN-LSTM (R3)	51.08%	0.516	0.345	+1.08pp
C3	AAPL	CNN-LSTM-Attn (R4)	54.98%	0.536	0.543	+4.98pp
C3	AMZN	CNN-LSTM-Attn (R4)	54.55%	0.559	0.492	+4.55pp
C3	GOOG	CNN-LSTM-Attn (R4)	52.81%	0.519	0.365	+2.81pp
C3	TSLA	CNN-LSTM-Attn (R4)	51.08%	0.433	0.367	+1.08pp

Table 5. Directional accuracy, ROC-AUC, and weighted F1 of the best model per condition and ticker (test set, n = 231 sequences).

5.3 comparison of all six architectures

The results presented in table 4 report mean directional accuracy across all tickers for each of the six architectures tested over the three conditions. The use of a single table allows for an easier direct comparison of the proposed CNN-LSTM models to all other classical and ablation baseline.

Model	C1 Mean	C2 Mean	C3 Mean	Overall
CNN-LSTM (Run 3)	51.08%	51.25%	50.98%	51.10%
CNN-LSTM-Attn (Run 4)	50.65%	49.18%	53.35%	51.06%
Random Forest	51.86%	50.65%	48.59%	50.37%
CNN-only	50.74%	49.87%	50.43%	50.35%
XGBoost	49.35%	50.13%	51.30%	50.26%
LSTM-only	49.35%	48.48%	50.87%	49.57%
Logistic Regression	47.45%	49.70%	49.35%	48.83%

Table 6. Mean directional accuracy across tickers for all six architectures. CNN-LSTM = Run 3; CNN-LSTM-Attn = Run 4.

5.4 Benchmark Comparison

The table below (table five) is a comparison of the directional accuracy of the two proposed models versus those that are provided in the current literature. The numbers for all of the benchmarks are representative of the average directional accuracy that was achieved by each of these methods using a multi stock data set and were subject to feature conditions as described with respect to C2.

Study / Model	C1	C2	C3	Evaluation Context
Fischer & Krauss (2018) — LSTM	—	53.20%	—	S&P 500, mixed regime
Nelson et al. (2017) — LSTM	—	55.90%	—	Individual stocks, bull
Lu et al. (2020) — CNN-LSTM	—	55.90%	—	Multi-stock, bull market
Sezer & Ozbayoglu (2018) — CNN	—	55.00%	—	DJIA constituents, bull
This study — CNN-LSTM (Run 3)	51.08%	51.25%	50.98%	2022 bear market
This study — CNN-LSTM-Attn (Run 4)	50.65%	49.18%	53.35%	2022 bear market

Table 7. Benchmark comparison. All figures are mean directional accuracy. Dashes indicate conditions not evaluated in the referenced study.

There is a very strong basis to believe that the bear-market conditions of the evaluation period, along with the use of a walk-forward validation approach that allows no overlapping training/testing sets, have reduced the potential maximum achievable accuracy. Therefore, achieving the same performance as reported by Fischer and Krauss in their full-data regime represents a positive outcome. Furthermore, since both studies employ essentially identical input representation techniques using publicly available data feeds, we can interpret the results as evidence of the effectiveness of the architecture under adverse data regimes.

5.5 Conditional Impact of Temporal Self-Attention

In contrast to expectations, our research indicates that there is no uniform benefit of incorporating self-temporal attention into CNN-LSTM architectures. Instead, the impact is highly dependent upon the composition of the feature-set:

C2 (Technical Features Only): The inclusion of self-temporal attention reduces the overall accuracy of the CNN-LSTM model by 2.07 percentage points (51.25% $\rightarrow$ 49.18%). In cases where all features within a particular dataset are comprised of high levels of inter-correlated information (i.e., each feature conveys similar trending information; e.g. the sixteen included technical indicators), self-temporal attention adds unnecessary complexity to the architecture. In such cases, it is attempting to isolate/distinguish timesteps that do not differ significantly from an informational perspective, thereby adding "noise" as opposed to "signal".

C3 (Technical Plus Sentiment Features): Self-temporal attention enhances the average accuracy of the CNN-LSTM model by 2.37 percentage points (50.98% $\rightarrow$ 53.35%). Sentiment differs qualitatively from technical indicators -- it is derived from an entirely separate source, displays differing rhythms, and on many occasions provides directional information that is orthogonal to price momentum. By learning to assign weights to those timesteps where sentiment is providing the greatest degree of useful directional information, self-temporal attention extracts a signal that would otherwise be smoothed away by the LSTM component.

This conditional relationship between feature diversity and the utility of self-temporal attention represents the most novel contribution to date of this study. Specifically, it appears that self-temporal attention provides greater utility under circumstances where the feature-set consists of diverse and sparsely-sampled signals of qualitatively different types. Conversely, when all features within a given dataset are highly correlated with one another, self-temporal attention is detrimental.

5.6 Statistical Significance—Diebold–Mariano Tests

To evaluate whether or not our results represent statistically-significant improvements in predictive performance, Diebold-Mariano (DM) tests with a small-sample correction developed by Harvey et al. (1997) were conducted to determine whether or not our proposed models represented statistically-significant superiorities over the five baseline models across all applicable tickers. Results from these tests are summarized below. Only one of 70 hypothesis tests produced a p-value $< .05$. That test compared MSFT C1 CNN-LSTM vs. XGBoost (DM = 2.571, $p = 0.011$), with the CNN-LSTM representing the superior model. Additionally, three tests resulted in p-values between .05 and .10: MSFT C1 CNN-LSTM vs. LR (DM = 1.719, $p = 0.087$); TSLA C1 CNN-LSTM vs. CNN-only (DM = -1.736, $p = 0.084$), with the CNN-only model superior; and GOOG C3 CNN-LSTM-Attn vs. LR (DM = 1.812, $p = 0.071$), with the CNN-LSTM-Attn model superior. The remainder of the tests did not produce statistically-significant results.

It should be noted that lack of widespread statistical significance in the DM-tests does not imply equivalence of performance amongst the competing models. Given that there are 231 test observations in this study, the DM-test possesses roughly 20 – 30 % power to detect a 2-percentage-point accuracy-difference between two models at the $\alpha = .05$ level. This implies that if a true accuracy-difference exists between two competing models and is as large as 2 – 3 percentage-points, then the probability that the DM-test detects this difference lies between 70 – 80%. Therefore, the primary limitation placed on this study's ability to demonstrate competitive predictive-performance through statistical inference is due to the shortness of the available sentiment data-window, not due to inadequacies in any of the proposed models.

Finally, confirmation of a single statistically-verifiable superiority of our proposed models (the MSFT C1 CNN-LSTM model vs. XGBoost) further supports the notion that our proposed models provide quantifiably superior predictive-performance when sufficient sample-power exists to detect such differences.

5.7 Explainability — Integrated Gradients Attribution

Integrated Gradients attribution was calculated for two different models across four tickers: (1) AAPL C2 CNN-LSTM and (2) AAPL C3 CNN-LSTM-Attn and three additional tickers.

For each ticker, attribution was calculated for every sequence in the test set (totaling 231 sequences), resulting in three-dimensional arrays of size (231, 20, n_features) — one attribution value per prediction, time-step, and feature.

5.7.1 Feature Importances — Technical Indicators (C2)

When averaged across all 231 predictions made by the AAPL C2 CNN-LSTM model, and all 20 time-steps associated with each prediction, revealed the relative importance of each feature within the overall attribution-profile. Of note, are that momentum-based oscillators provided higher degrees of attribution: RSI, MACD Histogram, and MACD were attributed higher values than any other feature in the case of AAPL. Price-based indicators — Open, High, Low, Close, EMA_20, EMA_50, Bollinger Bands — received lower attribution scores despite receiving much greater visual prominence in raw data feeds.

After applying Min/Max Scaling to the price-based features (to ensure they were comparable with technical indicators), all price-based features displayed high degrees of correlation with one-another (> 0.99 as previously documented during Data Understanding Phase) and conveyed primarily redundant trend information to the model. As a result, the model chose to focus on momentum-based oscillators (specifically those designed to identify directional momentum deviations).

5.7.2 Sentiment Attributions (C3)

Table 6 summarizes Integrated Gradients attribution values assigned to Sentiment across all four C3 tickers including Sentiment's ranking among all seventeen total features and whether or not its attribution-value had a positive or negative effect on predictions.

Ticker	Sentiment IG Value	Rank (of 17)	Direction	Interpretation
AAPL	0.002622	16th	Confirming (positive)	Positive sentiment → UP
AMZN	0.004151	17th (last)	Contrarian (negative)	Positive sentiment → DOWN
GOOG	0.002132	14th	Confirming (positive)	Positive sentiment → UP
TSLA	0.000521	17th (last)	Contrarian (negative)	Near-zero — uninformative

Table 8. Integrated Gradients attribution for the Sentiment feature in the CNN-LSTM-Attn C3 model (n = 231 test predictions per ticker).

In addition to technical indicators, sentiment was found to contribute some directional signal in the models. However, the inclusion of sentiment did not significantly improve results until an attention mechanism was incorporated into the models. This indicates that the model's ability to use sentiment effectively was largely dependent upon how well it could focus on relevant portions of the input. Overall, the model was found to be heavily reliant on technical indicators and used sentiment as a secondary indicator only. The directional effects of confirmation versus contrarianism also varied among tickers. For example, AAPL and GOOG benefited from having their respective sentiments confirmed (i.e., positive news predicting upward movement). On the other

hand, AMZN showed a contrarian relationship, such that positive news actually predicted downward movement during the 2022 bear market. Consistent with the broader macroeconomic picture at the time, these results suggest that even positive news regarding a specific company may have been insufficient to counteract the broad headwinds present in the markets at large. Finally, TSLA exhibited a near zero attribution (0.000521), suggesting that its overall sentiment remained relatively stable throughout the one-year testing period (i.e., standard deviation 0.057).

5.7.3 Temporal Attribution — What Days Matter Most

The temporal attribution profile for the AAPL C2 CNN-LSTM model clearly demonstrates a very strong recency bias. The most influential timestep was t-2 (two trading days ago) with a mean attribution of 0.030, closely followed by t-1 (yesterday) at 0.029. Attribution then declined smoothly toward older observations, with t-0 (yesterday's closing price) receiving a slightly lower attribution of 0.016 - suggesting the model discounts the most recent observation slightly, possibly due to end-of-day noise. The five most recent days together account for roughly 60 percent of total temporal attribution, while the ten oldest days account for only about 25 percent.

These results confirm that the model functions almost exclusively as a short-term momentum detection system. As such, it appears that the model is not learning either longer-term trends or cyclic relationships that occur within a 20-day window. Rather, it is extremely responsive to what happens in the past two to three trading days.

5.7.4 Attention Weights vs Integrated Gradients

Using the CNN-LSTM-Attn model applied to AAPL C3, normalized temporal attention weights were calculated using a simple average of all 231 test-prediction-based values. A Pearson correlation coefficient of 0.396 was then calculated between these attention-weighted profiles and the corresponding integrated gradients-based temporal profiles. While there is clear overlap between these two methods, they do not exhibit a strong degree of similarity.

It is useful to understand why there would be this partial alignment. Attention-weighted values represent learned temporal preferences during training that are identical for all subsequent predictions. Therefore, attention-weighted profiles capture what the model generally prefers in terms of temporal location throughout training. On the other hand, integrated gradients represent specific driving forces behind each unique test-prediction output. Thus, the fact that these two methods display moderate levels of correlation ($r=0.396$) supports that both methods measure actual temporal structure; however, since these two methods do not converge completely, they support that they can be viewed as complementary tools to provide a more complete understanding of temporal behavior.

5.7.5 Random Forest vs CNN-LSTM Feature Importance

A Random Forest model was trained on flattened versions of AAPL C2 sequences (the same training set used previously) with identical class weighting to that used previously and the resulting built-in impurity importances were summarized across the twenty time positions in order to derive feature-importance scores for each feature.

A Spearman rank-order correlation coefficient of $r=-0.744$ ($p\text{-value} < .001$) was then calculated comparing the Random Forest feature-importance scores and CNN-LSTM integrated gradients feature-importance scores. Furthermore, Random Forest ranked the features in almost perfectly inverse order compared to CNN-LSTM.

The highly negative correlation observed here provides perhaps the strongest evidence in this research that deep learning extracts fundamental differences in patterns in financial sequences compared to traditional tabular machine-learning methods. The Random Forest views the sequence as a flat table of 320 values and determines

which individual values (regardless of when they occurred) are positively correlated with UP or DOWN labels. In contrast, CNN-LSTM views the sequence as a series of temporal patterns -- i.e., sequences of feature values evolving over time -- that are positively correlated with UP or DOWN labels. Since these two types of information are nearly orthogonal in the AAPL C2 data, this supports the conclusion that the temporal structure of the 20-day window is primarily responsible for providing predictive signals and therefore that order-agnostic methods cannot gain access to this type of information.

Summary of Results

Overall, the analysis provided several key insights:

- 1. Architecture choice matters less than proper hyperparameter tuning.** By far the largest single accuracy improvement resulted from adjusting learning rate (from Run 0 to Run 1) rather than modifying architecture. Proper hyperparameters are essential prior to making valid comparisons of architectures.
- 2. Additional technical indicators added value.** Mean CNN-LSTM accuracy improved from 51.08% on C1 to 51.25% on C2 indicating that the eleven additional technical indicators contained additional directional information beyond pure OHLCV.
- 3. Sentiment added value only through attention.** CNN-LSTM on C3 performed at 50.98% without attention and therefore worse than C2 at 51.25%. Using attention on C3 increased accuracy to 53.35%, demonstrating that attention enables integrating sentiment.
- 4. Best result matched primary benchmark.** CNN-LSTM-Attn on C3 achieved 53.35% mean directional accuracy, matching Fischer and Krauss (2018) at 53.20% despite being evaluated over a much tougher bear-market interval.
- 5. GOOG always proved to be toughest ticker.** Regardless of condition or run, GOOG generated lowest accuracy values. GOOG had a significant contrarian relationship with sentiment ($r = -0.159$), and suffered extreme losses (-26.4% return and -34.9% maximum drawdown) in 2022 bear market.
- 6. Deep learning learns distinct patterns from data.** The substantial Spearman correlation ($r = -0.744$) between RF and CNN-LSTM feature-rankings indicated that deep learning sequential models learn patterns from temporal structure in sequences that classical ML models cannot access.
- 7. Model behaves as short term momentum detector.** Integrated Gradients temporal attribution indicate that most of predictive weight falls on final two days in 20-day window, with t-2 contributing most (mean attribution = 0.030).

6. Discussion and Insights

Section six transforms the quantitative Results of sections four and five into understandable language. It identifies how the recommended Solution addresses the initial issue, creates guidance for actions, and examines the constraints of the study so that you may understand those limitations before the actual deployment of your product.

6.1 What the Results Mean in Plain Business Terms

The central concern of this research was whether a deep learning model could reasonably predict if a stock will go up or down tomorrow relative to its price today. The honest response supported by all the data collected is yes -- however infrequently, weakly, and dependently.

A model using cnn-lstm with temporal attention using technical indicators and sentiment data as inputs produced the highest level of performance. During the period of september 2021 through september 2022, this model accurately forecasted the direction of five large-cap u.s. Technology stocks approximately 53.35% of the time. From a business perspective, a random guess (i.e., flipping a coin) produces an accurate guess 50% of the time. In other words, the model generates an accurate guess 3 to 4 times for each 100 guesses beyond random guessing alone. While the difference seems small, in terms of generating profitable trades for an algorithmic trader who executes dozens to hundreds of trades per day at very little expense per trade, the model's advantage represents a substantial amount of profit potential. Assuming that the model consistently generates profits over a period of time -- i.e., provided that the model does not suffer from "regression to the mean" due to random fluctuation -- the model can create significant economic value by virtue of a marginally higher edge than random guessing alone.

Two reasons make it particularly impressive that the model was able to work so effectively through an incredibly challenging time. The first reason is that the timing of this review was among the most treacherous trading conditions for technology stock performance ever recorded. In comparison, other tech giants including Amazon (down over 30%), Google (down over 26%), and Microsoft (down over 15%) fared worse. Even after experiencing a drop of almost 28% on one occasion before rebounding into ending approximately flat, Apple still outperformed those three by significant margins. The second factor is that the Federal Reserve raised interest rates at their fastest pace since 1980. This is why many models do very well in bull markets but ultimately will fail when faced with tough times like we just went through. Most of the time models are built to track or copy what's happening in a market that has momentum upwardly. They never attempt to predict where things are going.

6.2 How the Proposed Solution Addresses the Business Problem

As discussed in section one, there were two aspects to your business problems. The first issue was the lack of reliable tools for predicting directional movement in the equities markets that function under bear-market conditions. The second issue was the lack of a systematic way to combine sentiment data with technical signals into a single predictive model.

In regard to the first aspect, this project created a directional forecasting model that was specifically designed and tested on bear-market windows. As opposed to being programmed to merely follow along with bullish trends, this model is able to identify when the market is trending downward and correctly predict which direction that trend will continue. Therefore, for an investment manager, portfolio risk management team, or algorithmic trading group, a model that maintains performance regardless of which type of market exists is far superior to one that functions well only when the market is trending upward. Given that bear markets are more difficult environments than bull markets for any type of predictive model to function within, i believe that 53.35% is a reasonable minimum estimate of the capabilities of this model.

Regarding the second aspect, this project provides empirical support regarding how sentiment data should be incorporated into predictive models. If you add sentiment data as an additional input to a typical CNN-LSTM predictive model without any form of attention mechanism, you do not achieve enhanced predictive performance -- in fact, performance drops from 51.25% to 50.98% (without attention). However, incorporating sentiment into a predictive model only improves performance when a temporal attention mechanism is utilized in conjunction with a feature extraction/sequential memory component to determine which days' sentiment readings are most indicative of directional movement. Therefore, I recommend that organizations develop a design principle regarding deep learning-based forecasting models that incorporate sentiment data: unless an organization has reason to believe otherwise, they should assume that sentiment data is not useful until an attention mechanism is implemented in order to enable the model to select which days' sentiment readings are most relevant.

Finally, the explainable ai module in this project addressed one of the primary barriers to widespread organizational adoption of deep learning techniques in finance: the black box problem. An increasing number of regulators, risk managers, and members of investment committees require organizations utilizing automation in decision-making processes to provide explanations regarding how decisions are made. The integrated gradients attribution methodology applied in this project fulfills this requirement by providing for any prediction generated by the model a ranked listing of which input features contributed to the model's output and on which days those contributions were made. Therefore, instead of functioning solely as a generator of numbers that need to be interpreted by humans, this model now serves as an interpretive decision support tool whose rationale can be examined and potentially improved upon by humans.

6.3 Recommendations

I provide below Recommendations for any organization contemplating implementation of CNN-LSTM-based directional forecasting for technology equity portfolios based on my findings:

Recommendation 1: use the CNN-LSTM-attn model as only a directional signal input -- do not rely solely on it as a stand-alone trading system

While the directional accuracy obtained with this model (approximately 53%) represents a significant improvement over purely technical models (as evidenced by the comparison between the performance metrics presented in table four), it remains insufficient to serve as a stand-alone trading system that requires no human oversight. Instead, this model should be combined with other forms of input into a larger decision framework, e.g., signal dashboards, where human analysts can view both the direction predicted by this model and the associated confidence level for that prediction as well as see which input features (via integrated gradients attribution) contribute to that prediction. By combining human insight and judgment with this model's ability to consistently produce directional predictions with an edge greater than random guessing alone, organizations can capture this edge while maintaining oversight into unusual market events.

Recommendation 2: always incorporate sentiment data using temporal attention mechanisms -- never without them As demonstrated in this research (table six), simply appending sentiment data as another input feature to a cnn-lstm predictive model negatively impacts performance (50.98% vs 51.25%), indicating that incorporating sentiment as just another input feature is counter-productive. Organizations investing in sentiment data pipelines must therefore ensure that their chosen architecture incorporates a temporal self-attention mechanism between the cnn and lstm layers, allowing them to assign weights only to those days where sentiment carries directional information.

Recommendation 3: purchase higher-quality ticker-specific sentiment data

The sentiment data pipeline employed in this research (Kaggle sentiment dataset) has known quality issues: (a) AMZN and msft sentiment series exhibited identical values ($r=1$); (b) Tesla series contained nearly constant values throughout the year. Commercial financial NLP providers such as Bloomberg, Refinitiv or RavenPack generate high-frequency sentiment derived from news articles earnings call transcripts and social media at a level of detail that would likely yield substantially stronger more differentiated signals than those used here. The architecture is already developed to take advantage of better sentiment data; the primary constraint on development was quality of sentiment data - not design of model.

Recommendation 4: train the model to match the intended market environment

The walk-forward cross validation protocol developed in this study - including excluding the COVID crash fold from epoch selection vote - provides a template for regime-aware training. Prior to deployment in any production setting the model should be retrained with data ending as close to the current date as possible and the walk-forward validation should exclude all macro event periods that represent significant departures from historical norms (pandemic shocks, financial crises, extreme policy pivot events) from the epoch selection signal. A model trained on pre-2022 data and deployed in 2022 without re-training would have significantly underperformed.

Recommendation 5: concentrate deployment on stocks with rich differentially valuable sentiment coverage

The benefit realized by adding sentiment extension over technical model alone was most pronounced in stocks where sentiment provided meaningful information about direction (Apple: +1.73pp; Google: +6.06pp). For example, for Tesla which had near constant sentiment values and were virtually uninformative about direction the addition of sentiment produced minimal value. Organizations should evaluate both quality and variability of their sentiment data on a stock-by-stock basis prior to including sentiment into their models rather than assume that sentiment will provide benefits across all holdings equally.

Recommendation 6: apply the model to overnight position strategies – not medium-term portfolio allocation or long-horizon forecasting Integrated gradients temporal analysis leaves no doubt: the majority of predictive signal generated by the model derives from last two-to-three days trend patterns. Therefore the model is best suited for short term overnight position strategies and should not be applied to longer term portfolio allocations or trends forecasting horizons.

6.4 Limitations

The following limits of use represent the limitations of the current study and must be acknowledged by anyone using the findings of this study to make decisions.

Limitation 1: evaluation window. Models were tested on a single 252 day evaluation window running from september 2021 through september 2022. Although this window was chosen intentionally as a stress-test window, it remains a single window of measurement of model performance. It is possible that the same model could have produced very different performance during a different year - including years when there was a bull-run, correction, bear-market, or recovery. Therefore, an optimal testing procedure would involve testing over multiple non-overlapping test windows across a variety of market regimes, i.e., bull-runs, corrections, bear-markets, recoveries, etc. This would provide a more complete assessment of whether or not the directional edge of the model is robust across changes in market regimes.

Limitation 2: quality of Sentiment data. There are three known quality issues with the free Kaggle Sentiment dataset used in this study. First, both amazon and Microsoft Sentiment series share a perfect correlation ($r = 1.0$) which necessitated excluding Microsoft from the analysis. Second, Sentiment regarding tesla exhibited virtually no variation throughout the test year and therefore provided little useful information. Third, Google's Sentiment series contained fourteen missing values requiring imputation. Due to these quality issues, the conclusions drawn from this study regarding the utility of Sentiment are based upon a lower quality signal than what is typically available. As such, if higher quality Sentiment signals were available, it is probable that the results would demonstrate significantly better performance and potentially greater explanatory power. Therefore, caution should be exercised when attempting to generalize the results of this study related to Sentiment beyond the specific dataset employed in this study.

Limitation 3: sample size and statistical power. Based on a sample size of 231 test sequences, the Diebold-mariano test has only 20% to 30% statistical power to identify an actual two percentage points accuracy difference at a Five percent significance level. Additionally, Only one out of seventy statistically significant tests demonstrated a positive result. The above does not necessarily indicate that the performance differences did not exist; however, it indicates that the evaluation window employed in this study was insufficiently long to reliably measure such differences. Organizations employing this model should consider the reported accuracy measures as point estimates with considerable uncertainty associated with such estimates, rather than precise measurements.

Limitation 4: limited number of stocks studied. This study analyzed only Five u.s.-traded technology stocks. Specifically, all Five are large-cap stocks, highly liquid and widely followed by analysts and social media. Consequently, the findings may not be applicable to mid-cap stocks, international equities, less liquid securities or other sectors such as energy, health care, financials where stock price behavior, Sentiment influences, and speed of information dissemination differ from those found in the technology sector.

Limitation 5: costs not incorporated. The accuracy percentages presented are prior to accounting for Transaction costs and market impact. In reality, a trading strategy implementing trades on the basis of this predictive model will incur costs on every trade that reduce the profit earned per trade and thereby erodes some portion of the three-percentage-point edge. Although Transaction costs are generally low for high liquidity large-capitalization stocks similar to those included in this study due to their high liquidity levels; however, they do exist. Thus, if this model is converted into a live trading system, then all-in costs need to be carefully accounted for before concluding that the edge is economically viable.

Limitation 6: Training data does not include test year Sentiment signal. Since Sentiment data only existed for the test year (i.e., 2021 – 2022), the Condition C3 models were trained with Sentiment equal to zero for all Training examples. Real-time Sentiment was only available during testing. Hence, the model had limited opportunities to develop knowledge regarding how to correlate Sentiment values with future price movement during Training. If a longer history of Sentiment data was available (i.e., covering the entire Training period from 2015 – 2021), then the model would have been able to develop knowledge regarding the correlation between Sentiment values and future price movement under many more varied market conditions than currently available. Therefore, it is anticipated that Condition C3 performance would be improved.

Limitation 7: single random seed used for model initializations. All models were initialized using seed = 42, a commonly employed technique for ensuring reproducibility. However, initializing models using a single random seed can produce results that are slightly different when trained using different initialization conditions. Employing a more rigorous methodology would require Training each model configuration Five to ten times using different random seeds and reporting mean accuracy along with standard deviation to establish a

confidence interval around each point estimate instead of relying on a single estimate. The walk forward cross validation methodology used in this study provides some mitigation against this limitation; however, it does not completely eliminate concerns regarding multiple model initializations.

6.5 Conclusion

This project was established to explore if a CNN-LSTM hybrid deep-learning model, enhanced with temporal self-attentive layers and improved using sentiment as additional features, can provide consistent directionally predictive next day forecasts for large-cap U.S. technology companies during periods of very high volatility in the markets. The results suggest we should cautiously say "yes".

The CNN-LSTM-Attn Model based on technical indicators and using sentiment data produced an average directional forecast accuracy of 53.35 % over the 2022 bear-market evaluation period. This result matches the academic benchmark (Fischer & Krauss, 2018) but was generated in much harder conditions than those used in Fischer & Krauss (2018). Our CNN-LSTM-Attn Model performed better than our classical baseline models (Logistic Regression, Random Forest, XGBoost), as well as our deep learning ablations (CNN-Only, LSTM-Only) when analyzing the sentiment-enriched condition. Each of the design decisions made in building this architecture (combining local feature extraction, sequential memory, temporal attention, and integrating features from different sources) were supported and proved useful together.

The Explainable-AI analysis demonstrated that our model is fundamentally a short-term momentum predictor and relies almost exclusively on the last 2 to 3 trading days and momentum oscillators rather than price levels or moving averages. The sentiment input only contributed to the forecasting when combined with the use of temporal attention. We found that the contribution of the sentiment varied by ticker and had a clear economic interpretation. The negative Spearman correlation of $r = -0.744$ between Random Forest and CNN-LSTM feature ranking further confirmed that deep learning with sequence can access different information from the same data than traditional machine learning can access.

This research study provides insights into how data analysts and software developers designing information systems can create accurate predictive models. In particular, it shows that the details of the training process (learning rate, model size, validation approach and selecting epochs aware of current regime) contribute more to the eventual accuracy of the predictions than does individual architectural design decisions. Additionally, this study illustrates that transparency of model behavior using attribution techniques can be both technically possible and socially needed before releasing such models into production environments for making financial decisions.

The limitations of this study (single evaluation window, poor quality sentiment data, relatively small test samples, focus on just five equities) clearly indicate areas where future research will need to expand upon the conclusions drawn here. Future studies expanding this research's empirical scope to include other market regimes, incorporating professional-quality sentiment data and including larger numbers of equities would greatly increase the generality of these findings. The methodology and associated code developed in this research project are structured to allow for modularity and extensibility; thus establishing a reproducible base for such expansions.

7.Future Research Directions

These results also create a number of potential avenues for additional research that will significantly enhance our understanding of the conclusions we have reached in this paper.

The first step in this process is to evaluate the model under various market conditions as opposed to a single (bear) market condition. By testing the model over an entire economic cycle (including bull markets, corrections, bear markets and recoveries), we can make a better determination of whether the directional advantage exhibited by the model is consistent across all market environments or unique to the 2022 rate hike period.

Secondly, there are improvements to be made in terms of the quality of the sentiment data used. As previously discussed, the free Kaggle data source was limited in its quality. A professional grade sentiment data source such as those provided by Bloomberg, Refinitiv, or RavenPack; which provide high frequency ticker-specific signals based on earnings calls, analyst reports and news articles; will enable us to perform a more thorough evaluation of whether sentiment provides meaningful forecasting power when the signal is clean and well defined.

Thirdly, it would be beneficial to expand upon the study using a larger universe of stocks. Utilizing the same methods and analysis techniques employed within this paper but expanding to include mid cap companies, international equity securities, and companies classified in different industries (such as energy, health care and financials); will allow us to assess how broadly applicable the findings from this study are relative to large capitalization U.S. technology equities.

Lastly, future research should investigate and compare more complex forms of attention. The single head additive attention mechanism utilized in this study represents one form of attention. Other forms of attention may include multi-head self-attention, cross-attention between sentiment and price sequences or even completely replacing the LSTMs with a transformer architecture. These more advanced forms of attention may extract richer patterns from the same input sequence of stock prices and sentiment information.

Disclaimer

This report was created as a project for academic purposes in fulfillment of the requirements of the Capstone Research Project (MATH-619) which is offered to students enrolled in the Mathematics Department at King Fahd University of Petroleum and Minerals. The report serves only as an example of how one might approach conducting research and demonstrating analytical capabilities in the fields of data science and financial forecasting.

Therefore, nothing herein constitutes financial advice, investment guidance, nor should it be construed as recommending to buy, sell or hold any security or financial instrument. In addition, all predictive models, conclusions drawn from analysis, results of analyses and recommendations made in this report are the result of an academic research endeavor and have neither been designed, tested nor validated for use with actual securities, or in real-time trading environments.

Additionally, all past performance comparisons shown in this paper represent historical data and were obtained over a very narrow time frame; therefore, these studies may not provide a realistic representation of current or future market conditions. As stated earlier, markets are highly volatile due to many factors including changes in government policies, geopolitical events etc., which cannot be predicted accurately using any model. Therefore, even if a model has consistently demonstrated its ability to predict past returns, there is no assurance that such consistency will continue into the future. The time interval chosen for the test in this study (September 2021 – September 2022), was selected to represent one particular and unique market environment and may not reflect what could occur under different market conditions.

Furthermore, the models developed during the course of this research were not subjected to the same level of scrutiny that would normally be applied prior to implementing any model for the purpose of generating trades in any commercial or institutional trading setting. Therefore, any person or organization choosing to take action on the information, models and/or recommendations provided in this report do so entirely at their own peril.

The Author will assume no liability whatever for any loss, damage, or negative consequence(s), whether direct or indirect, resulting from reliance on or misuse of the material contained in this report.

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